BIKE STATION SHARING DASHBOARD

1. OVERVIEW

This Power BI dashboard presents an analytical overview of a bike station sharing system across multiple cities. The objective is to analyze station distribution, bike availability, city-wise analysis, station availability, operational status, and category wise analysis using interactive visualizations.

The dataset includes station ID, station name, address, city, latitude, longitude, total bike stands, available bikes, available bike stands, station condition, banking, bonus, status, and timestamp information.

2. DATA CLEANING PROCESS

The dataset was cleaned and prepared using Power Query Editor in Power BI with the following steps:

- Standardized column names for better readability
- Split position column into **Latitude** and **Longitude**
- Converted data types (text, numeric, and datetime formats)
- Replaced null values in Station address to Unknown
- City column is formatted to Capitalize each word
- Created calculated columns such as:
    - Empty Status – Available bikes =0 empty/Non-Empty
    - Station Condition – Empty, full, low bikes, high bikes
    - Is Latest – Max date (Latest=1)

3. OBJECTIVES OF THE DASHBOARD

1. **Station Distribution Analysis** – Understand spatial distribution of bike stations
2. **City-wise Performance Analysis** – Compares bikes and stands availability across cities
3. **Stands Availability Analysis -** Analyze how availability percentage varies across different bike stands

4. **Top 10 Stations by Bike Availability** -Ranking Top 10 bike stations with the highest number of available bikes
5. **Banking Feature on Bike Availability** – Examine bike availability percentage based on banking
6. **Operational Status Monitoring** – To analyze how many stations are open, closed
7. **Station Condition Analysis –** Number of stations grouped by the station conditions like balanced, low bikes, high bikes, empty and full

4. VISUALIZATIONS USED

- KPI Cards -> (Total Stations, Total Bike Availability, Total Bike Capacity, Availability %, Empty Stations, Full Stations)
- Map Visual -> (Bike Station Distribution)
- Bar Chart -> (City wise availability of bikes and stands)
- Tree Map -> (Availability % by bike stands)
- Column Chart -> (Top 10 Stations by Bike Availability)
- Pie Chart -> (Availability % by banking)
- Funnel Chart -> (Station count by status)
- Donut Chart -> (Station count by Station condition)

5. DAX MEASURES

- Total Stations
- Total Bike Availability
- Total Bike Capacity
- Availability %
- Empty Stations
- Full Stations
- Station Count
- Occupancy %
- Average bike per station

5. KEY INSIGHTS

1. **Bike Station Distribution-** The map visual shows the geographic distribution of bike stations, where bubble size represents bike availability and color (legend) represents station condition to identify usage patterns across locations.

2**. City wise Bike and stands availability-**

- **Bruxelles-Capitale** – High bike availability (3532) means good supply
- **Lyon** – More stands (9350) means bigger network, built for high demand

3. **Stands Availability**

- More stands in the station 10002-INSA (55)

4. **Top 10 stations**

- The column chart highlights the top 10 bike stations with the highest bike availability, indicating stations with either high supply or low demand.

5. **Availability by Banking**

- Most stations are non-banking enabled, shows that users mainly use non-banking stations for bike sharing activity (Banking=False 63.47%)

6. **Operational Status**

- More stations show open status (936)
- Closed stations (92)

7. Station condition

Empty stations (295) are more says that no available bikes, indicating full utilization or complete unavailability

6. CONCLUSION

The bike station sharing dashboard provides a clear understanding of station distribution, availability patterns, and operational performance across different cities. The analysis shows that cities like Bruxelles-Capitale and Lyon have strong infrastructure with high bike availability and larger station capacity, indicating well-developed networks. However, a significant number of stations are found to be empty or fully utilized, highlighting imbalances in bike distribution and demand.

Most stations are operational and open, showing good system availability, but the presence of empty stations and variation in availability across locations suggests the need for better bike redistribution strategies. Additionally, non-banking stations dominate usage, indicating that users primarily rely on standard bike-sharing access rather than banking-enabled services.

Overall, the system is functional and widely used, but improvements in bike allocation and balancing between stations can enhance efficiency and user experience.